EXPERIMENT – 09

AIM:

To develop an Android application that implements a basic calculator using Android controls such as Button, TextView, and EditText. The calculator will support basic mathematical operations like Addition, Subtraction, Multiplication, and Division.

ALGORITHM:

1. Initialize the Android Application:

o Set up a new Android project in Android Studio.

o Create a MainActivity Java or Kotlin file for the main screen of the application.

2. Design the User Interface (UI):

o Create the following UI components in the XML layout file:

EditText for input fields where the user will enter numbers (two EditTexts for number1 and number2).

Buttons for each arithmetic operation (Addition, Subtraction, Multiplication, Division).

TextView to display the result of the operation.

3. Declare Variables:

o Create variables for the EditText fields, Buttons, and TextView.

o Define variables to hold the input numbers and the result (e.g., num1, num2, result).

4. Get Input from the User:

o Get the values entered by the user in EditText fields using getText().toString() and convert them to numerical values (like Integer.parseInt() or Double.parseDouble()).

5. Set Event Listeners for Buttons:

o For each button (Addition, Subtraction, Multiplication, and Division), set an OnClickListener.

o Inside the onClick() method for each button, perform the respective arithmetic operation:

Addition: result = num1 + num2

Subtraction: result = num1 - num2

Multiplication: result = num1 * num2

Division: result = num1 / num2 (add validation to prevent division by

zero).

6. Perform Arithmetic Operation:

o Check which button was clicked (using view.getId()), and based on the button, perform the respective arithmetic operation.

o Handle Division by Zero: Check if num2 is 0 before performing division and show an appropriate error message if true.

7. Display the Result:

o Use TextView to display the result of the operation.

o Convert the result to a string and set it using setText() method on the TextView.

8. Validate User Inputs:

o Before performing any operation, validate the input to ensure the user has entered valid numbers (not empty or non-numeric characters).

9. Handle Edge Cases:

o Ensure the app handles cases where users input invalid values or attempt division by zero. Display appropriate messages for errors.

10. Finish the Operation:

o Once the result is displayed, the user can either perform another calculation or exit the app.

PROGRAM:

Step 1: Create the Project

Template: Empty Views Activity

Project name – CalculatorApp

Package - com.example.calculatorapp

Language: Java

Minimum SDK – API 24

Build system: Groovy DSL

Step 2:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
android:orientation="vertical"
android:background="#1a1a2e"
android:padding="20dp">

<!-- Header -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="🏠 Hostel Management"
    android:textColor="#4CAF50"
    android:textSize="20sp"
    android:textStyle="bold"
    android:gravity="center"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Fee Calculator"
    android:textColor="#aaaaaa"
    android:textSize="13sp"
    android:gravity="center"
    android:paddingBottom="20dp"/>

<!-- Input 1 -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="First Value"
    android:textColor="#dddddd"
    android:textSize="14sp"
```

```
    android:paddingBottom="5dp"/>

<EditText
    android:id="@+id/etNum1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter first number"
    android:inputType="numberDecimal"
    android:backgroundTint="#4CAF50"
    android:textColor="#ffffff"
    android:textColorHint="#888888"
    android:paddingBottom="15dp"/>

<!-- Input 2 -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Second Value"
    android:textColor="#dddddd"
    android:textSize="14sp"
    android:paddingBottom="5dp"/>

<EditText
    android:id="@+id/etNum2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter second number"
    android:inputType="numberDecimal"
    android:backgroundTint="#4CAF50"
    android:textColor="#ffffff"
    android:textColorHint="#888888"
```

```
    android:paddingBottom="20dp"/>

<!-- Buttons Row 1: Add and Subtract -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="12dp">

    <Button
        android:id="@+id/btnAdd"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="+ Add"
        android:backgroundTint="#4CAF50"
        android:textColor="#ffffff"
        android:layout_marginEnd="8dp"/>

    <Button
        android:id="@+id/btnSub"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="− Subtract"
        android:backgroundTint="#2196F3"
        android:textColor="#ffffff"
        android:layout_marginStart="8dp"/>

</LinearLayout>
```

```
<!-- Buttons Row 2: Multiply and Divide -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="20dp">

    <Button
        android:id="@+id/btnMul"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="× Multiply"
        android:backgroundTint="#FF9800"
        android:textColor="#ffffff"
        android:layout_marginEnd="8dp"/>

    <Button
        android:id="@+id/btnDiv"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="÷ Divide"
        android:backgroundTint="#e53935"
        android:textColor="#ffffff"
        android:layout_marginStart="8dp"/>

</LinearLayout>

<!-- Result Box -->
<LinearLayout
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:background="#2e2e4e"
    android:padding="20dp"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result"
        android:textColor="#aaaaaa"
        android:textSize="13sp"
        android:paddingBottom="8dp"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="—"
        android:textColor="#4CAF50"
        android:textSize="28sp"
        android:textStyle="bold"/>

</LinearLayout>

<!-- Clear Button -->
<Button
    android:id="@+id/btnClear"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
        android:text="Clear"
        android:backgroundTint="#555555"
        android:textColor="#ffffff"
        android:layout_marginTop="15dp"/>

</LinearLayout>
```

Step 3:
Main_Activity.java:

```
package com.hostel.management;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etNum1, etNum2;
    TextView tvResult;
    Button btnAdd, btnSub, btnMul, btnDiv, btnClear;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Link XML views to Java variables
```

```
etNum1    = findViewById(R.id.etNum1);
etNum2    = findViewById(R.id.etNum2);
tvResult  = findViewById(R.id.tvResult);
btnAdd    = findViewById(R.id.btnAdd);
btnSub    = findViewById(R.id.btnSub);
btnMul    = findViewById(R.id.btnMul);
btnDiv    = findViewById(R.id.btnDiv);
btnClear  = findViewById(R.id.btnClear);

btnAdd.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        double[] nums = getInputs();
        if (nums == null) return;
        double result = nums[0] + nums[1];
        tvResult.setText(String.valueOf(result));
    }
});

btnSub.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        double[] nums = getInputs();
        if (nums == null) return;
        double result = nums[0] - nums[1];
        tvResult.setText(String.valueOf(result));
    }
});

btnMul.setOnClickListener(new View.OnClickListener() {
    @Override
```

```
    public void onClick(View v) {
        double[] nums = getInputs();
        if (nums == null) return;
        double result = nums[0] * nums[1];
        tvResult.setText(String.valueOf(result));
    }
});

btnDiv.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        double[] nums = getInputs();
        if (nums == null) return;
        if (nums[1] == 0) {
            Toast.makeText(MainActivity.this,
                "Cannot divide by zero", Toast.LENGTH_SHORT).show();
            return;
        }
        double result = nums[0] / nums[1];
        tvResult.setText(String.valueOf(result));
    }
});

btnClear.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        etNum1.setText("");
        etNum2.setText("");
        tvResult.setText("—");
    }
});
```

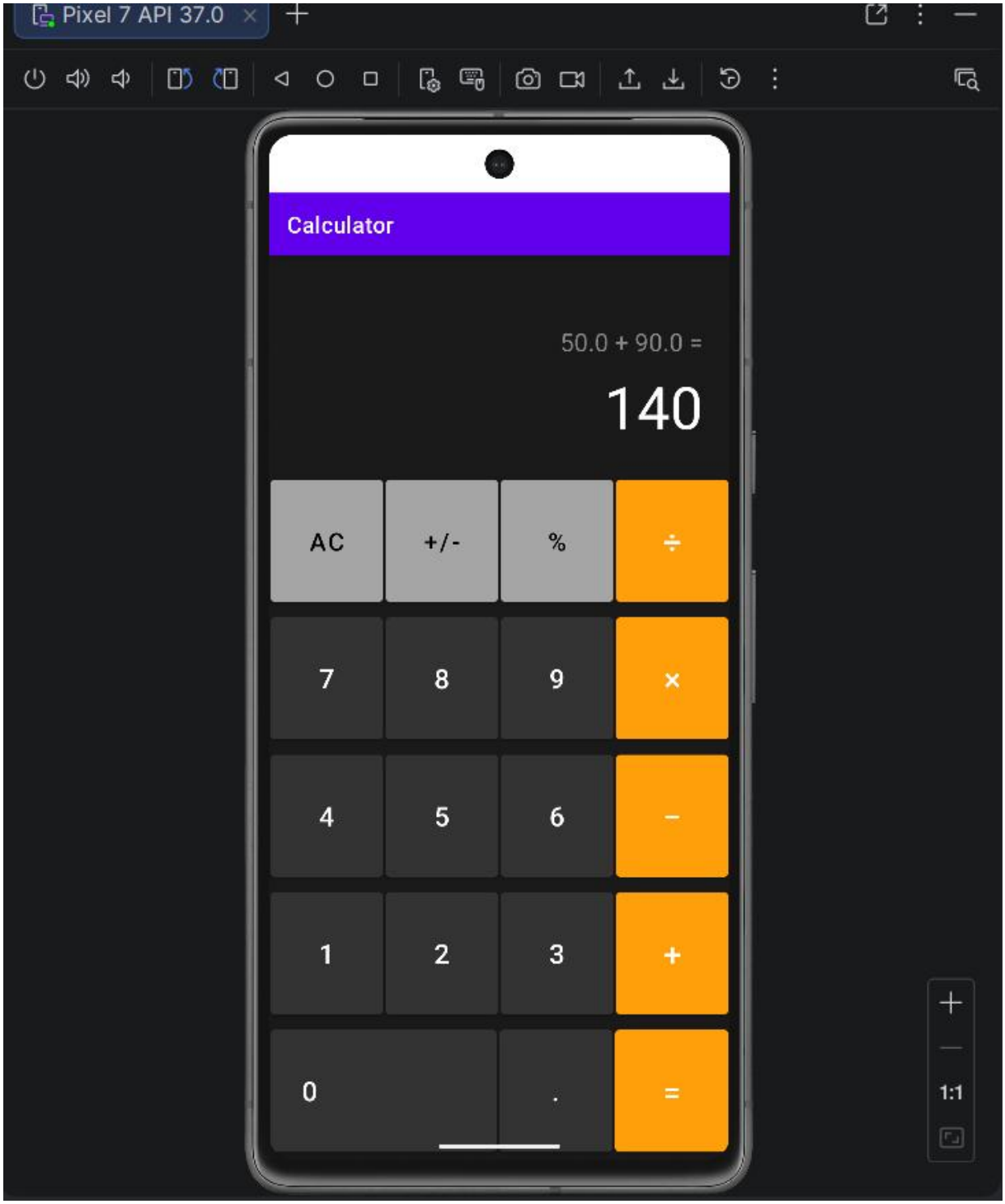
```
    }

    // Helper: reads and validates both inputs
    private double[] getInputs() {
        String s1 = etNum1.getText().toString().trim();
        String s2 = etNum2.getText().toString().trim();

        if (s1.isEmpty() || s2.isEmpty()) {
            Toast.makeText(this,
                "Please enter both values", Toast.LENGTH_SHORT).show();
            return null;
        }

        double n1 = Double.parseDouble(s1);
        double n2 = Double.parseDouble(s2);
        return new double[]{n1, n2};
    }
}
```

OUTPUT:



Result: A calculator is designed using Android studio which uses controls like Button, TextView, EditText having basic functionality like Addition, Subtraction, multiplication, and Division.